Siddharth Paul

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GitHub | LinkedIn | Portfolio

Education

Chandigarh University — Mohali, Punjab, India
Bachelor of Engineering in Computer Science and Engineering; CGPA: 7.99/10 — *Jun 2024 – Expected May 2028*

- **Coursework:** Data Structures & Algorithms, Operating Systems, Computer Networks, DBMS, Computer Architecture, System Design, Competitive Coding

Projects

DiskTree – Disk Usage Analyzer | *C++20, CMake, FTXUI, Catch2* | GitHub — May 2026 – Jun 2026

- Built a multithreaded disk-usage analyzer in C++ that indexes **230K files across 210 GB in ∼10 s**, using an iterative (stack-based) DFS scanner, avoiding recursion-depth limits on deep directory trees
- Parallelized traversal with a fixed-size thread pool, dispatching directory subtrees to worker threads as tasks instead of spawning a thread per directory
- Wrote a bump-pointer arena allocator for tree nodes, replacing hundreds of thousands of per-node heap allocations with pointer bumps and releasing the tree by freeing a few large blocks
- Designed a scan → build → index pipeline with bottom-up size roll-ups and a pre-sorted flat index so **top-N queries need no re-scan**, shared by three front-ends: a CLI, an interactive FTXUI terminal UI, and a local web UI (embedded HTTP server + D3 treemap)
- Added JSON/CSV export and zstd-compressed snapshots with scan-to-scan diffing to track disk growth over time; shipped a zero-dependency single-binary release with Catch2 unit tests and GitHub Actions CI

Voice-First Multilingual RAG System | *Python, FastAPI, BM25, HNSW* | Demo | GitHub — Aug 2026

- **Led a 3-person team**: architected the end-to-end system (speech input → retrieval → answer extraction → API → web UI), split it into modules and assigned them, and personally built the retrieval core
- Implemented hybrid retrieval over **ai4bharat/MSMARCO-XI**, fusing sparse BM25 with dense `multilingual-e5-small` embeddings in an HNSW vector index via Reciprocal Rank Fusion, enabling cross-lingual search for native-script queries (Hindi, Bengali)
- Replaced LLM generation with extractive span selection from retrieved passages, keeping end-to-end latency **under 200 ms** with per-stage latency instrumentation
- Exposed the pipeline through a FastAPI backend and deployed it with a web front-end for live voice queries

Multi-Agent Orchestration Daemon | *Go, Python, gRPC, Protocol Buffers* | GitHub — Feb 2026 – Present

- Designing and building a local-first, always-on Go daemon that supervises polyglot AI agents (Python/Go) as independent OS processes, isolating agent crashes from the control plane
- Defined strongly typed gRPC/Protobuf contracts for agent registration, task dispatch and server-streamed logs, with generated Go and Python stubs so agents in either language share one interface
- Building heartbeat-based failure detection and a task dispatcher on top of the agent registry (in progress)

CP Companion | *Rust, Tauri v2, React, TypeScript, SQLite* | Demo | GitHub — Jun 2025 – Aug 2025

- Built a cross-platform desktop app in Rust/Tauri that aggregates upcoming contests from Codeforces, LeetCode, CodeChef and AtCoder via the Clist API, with calendar and list views
- Cached API results in SQLite behind a Rust backend served over Tauri IPC, so a second always-on-top countdown widget window syncs from the local cache instead of re-hitting the rate-limited API

Technical Skills

Languages: C++, Go, Java, C, Rust, Python, TypeScript, SQL, Bash
Systems: Multithreading, memory management, Linux, gRPC/Protobuf
Databases: PostgreSQL, MySQL, MongoDB, SQLite
Tools: Git, Docker, CMake, GitHub Actions, Catch2
Frameworks: Spring Boot, Node.js, Express, FastAPI, React

Achievements & Certifications

Smart India Hackathon: won the internal round at Chandigarh University twice in a row (2025, 2026)
Machine Learning Specialization – Stanford Online & DeepLearning.AI (Coursera, 2026) | Credential